

## Data Input Layer

Source Domain  $D_S$   
Normal Weather  
( $n_S$  samples)

Distribution Shift

Target Domain  $D_T$   
Extreme Weather  
( $n_T \ll n_S$ )

## Feature Mapping

Kernel  $\phi : \mathcal{X} \rightarrow \mathcal{H}$   
(RBF / Nyström)

Complexity:  $O(N^3) \rightarrow O(TNm)$

## Joint Optimization $\min_{\mathbf{w}, b} J(\mathbf{w}, b)$

Structural Risk  
 $\frac{1}{2} \|\mathbf{w}\|^2$

+

Variance (LDMR)  
 $\lambda_1 \sum (u_i - \bar{u})^2$

+

MMD Adaptation  
 $\lambda_2 \text{MMD}^2(D_S, D_T)$

+

Pinball Loss  
 $C_S \sum L_\tau + C_T \sum L_\tau$

$C_T \gg C_S$

## Probabilistic Forecasting Output

Optimal Model  
 $f(\mathbf{x}) = \mathbf{w}^T \phi(\mathbf{x}) + b$

Upper:  $\hat{q}_{0.95}$

Median:  $\hat{q}_{0.50}$

Lower:  $\hat{q}_{0.05}$

} PI